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# Interference analysis for asynchrounous OFDM/FBMC based cognitive radio networks over Rayleigh fading channel

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**Abstract**
In the future fifth generation networked society, devices will suffer from the asynchronous multi-carrier effect which impacts the user's quality of experience. This paper investigates the timing misalignment effect on interference level in the context of a cognitive radio (CR) network. Our study considers both multicarrier techniques orthogonal frequency division multiplexing (OFDM) and filter bank multicarrier (FBMC). The originality of our paper consists in proposing a simple but accurate analytical model to evaluate our system's performance in terms of interference level, signal-to-interference-plus-noise ratio and bit error rate. Specifically, two case studies are considered, a single-user case (one primary user and one secondary user) and a multi-user case with exact theoretical expressions of interference level. We had also made a comparison between OFDM and FBMC techniques. The most striking observation to emerge from our results is that the asynchronous interference is inversely propositional to timing offset even for the multi-user case, and that differences in the normalized interference level between the single/multi-user cases is barely perceived for the FBMC technique, proving its efficiency. Our simulations' results had further strengthened our confidence in the suggested model. It shows that FBMC is best suited for CR networks since it provides BER improvements compared to OFDM.

**Keywords** Interference · Cognitive radio · OFDM · FBMC · Asynchronous · Analytical approach

## 1 Introduction

Fundamental research for fifth generation (5G) networks aims primarily to improve data rate and coverage. The 5G will not only be an evolutionary solution for mobile network but it will also offer a new network capability and sophisticated broadband services based on the integration of the wired and the wireless world. It will be the key of Internet of Things (IoT) [1] aiming to offer a 1000× network capacity and 10× energy efficiency. So what are the fundamentals of 5G and what are the technical challenges facing its deployment? In fact, interference evaluation is considered as an important issue causing degradation for QoS especially in cognitive radio (CR) networks [2]. Since the multi-carrier system performance suffers in some cases from degradation that is due to imperfect time and frequency synchronization, many works are dealing with asynchronous interference. Therefore, for cognitive radio, non cooperative base stations, sensor networks [3] or ad-hoc networks, it is very difficult to have perfect synchronization. The lack and underutilization of spectrum has led to the development of CR technology, which exploits the existing spectrum opportunistically by dynamic spectrum access (DSA) [4, 5]. Therefore, it can be defined as a system where unlicensed or secondary user (SU) can temporarily communicate over unused spectral bands, the licensed or primary user (PU) are in reality their owners [6]. As a function of radio conditions, the awareness of the surrounding radio environment is essential to enabling cognitive devices to react, adapt, and eventually optimize resource usage and performance. In fact, different models have been proposed to investigate the issue of interference such as the classical Gaussian approximation (GA) techniques [7], and the power spectral density (PSD) modeling

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[8]. In our work, we are using instantaneous interference tables in order to model asynchronism in multi-carrier CR networks. In [9], a new interference model for cognitive radio networks was proposed for two multi-carrier modulation techniques: OFDM and FBMC. Our analytic approach was developed for a single-user case, where we consider only one PU and one SU. Furthermore, it is well known that a spatial spectrum sharing could be introduced for massive MIMO based CR system which is considered as a promising technology in 5G networks [10]. Another crucial issue related to the interference analysis topic, that is drawing intensive attention, is the cognitive beamforming (CB). In the literature [11], CB often refers to the fact of making a tradeoff between the maximization of the throughput and the interference avoidance.

Consequently, this paper studies the asynchronous interference effect on the system's performance. Therefore, a theoretical aspect of the asynchronous interference within a cognitive radio (CR) network is presented. Our goal is to analyze the asynchronous interference for CR network. We notice also that we model the interference by an instantaneous interference table for each timing offset for a multi-user case (N PUs and N SUs). In this work, we will characterize and analyse the suggested asynchronous interference modeling for the 5G CR network by using two multi-carrier techniques which are OFDM and FBMC.

Following the introduction, the related works are presented in (Sect. 2). The first part of this paper (Sects. 3, 4) focuses on a brief overview of the system model and a theoretical background of the suggested asynchronous interference analysis. Numerical results are then detailed in Sect. 5 focusing on both OFDM and FBMC multicarrier techniques. Finally, Sect. 6 presents the conclusion to this paper.

## 2 Literature review

To answer the challenge of next generation of wireless networks (5G), important research efforts have been established in order to model and evaluate the effect of asynchronism on the system's performance. To the best of the authors' knowledge, asynchronous interference can be quantified with the power spectral density (PSD)-based model as in [12]. The PSD is generally used to evaluate the mutual interference between two systems, however it does not consider the real level of interference since the asynchronism is absorbed. As interference optimization is a significant issue for better QoS, the impact of asynchronism, specially for different OFDM and FBMC techniques, was investigated in literature [13, 14]. Previous works dealing with asynchronous interference are generally in the context of wireless networks without considering the CR context. Indeed, timing asynchrony resulting in timing offset errors between the emitter and the receiver have relevant effect for CR networks since the coexistence between SU and PU must be with the best performance for each of them. Since the PU/SU are using multi-carrier waveforms and transmitting independently, the mutual asynchronous interference is more considerable. In addition, by scanning preliminary work in this field, related research focused primarily on the impact of multiple PUs and fading channel on the spectrum sensing performance as in [15]. Many other methods, based on the interference alignement (IA), have been proposed [16, 17] in order to deal with the interference issue in CR networks. IA relies on the coordination between transceiver pairs by aligning interference signals in space time and frequency. It is considered as a challenging area in the field of CR for interference management and it shows that at high SNR, the sum rate can scale linearly with the number of users. Since our focus is on asynchronous interference, the IA could be used in asynchronous multi-carrier based CR networks, it would lessen the interference existence.

## 3 System model

In wireless communication, interference is considered as the main cause leading to quality degradation of the system's quality of service. In this study, we suggest a global analytical model of the interference in asynchronous multi-carrier systems in the CR context. This type of interference is the result of orthogonality loss between different sub-carriers. In considering the transmission in OFDM/FBMC based multi-cellular networks, interference analysis is related to asynchronous effect in frequency selective fading channels.

The nodes can be either licensed or unlicensed users. Accordingly the system consists of PUs and SUs, as depicted in Fig. 1.

In this paper, we address the effect of timing misalignment and its consequence in the context of multi-carrier networks. The system performance will be addressed in terms of asynchronous interference, signal-to-interference-plus-noise ratio (SINR) and power.

Thus, our goal is to:

(1) Estimate the interference level in the presence of timing misalignment for both OFDM and FBMC CR networks with the single-user case for a limited number of PUs and SUs (one PU, one SU);
(2) Extend the performance analysis to the multi-user case, since it has not been considered in the literature before;

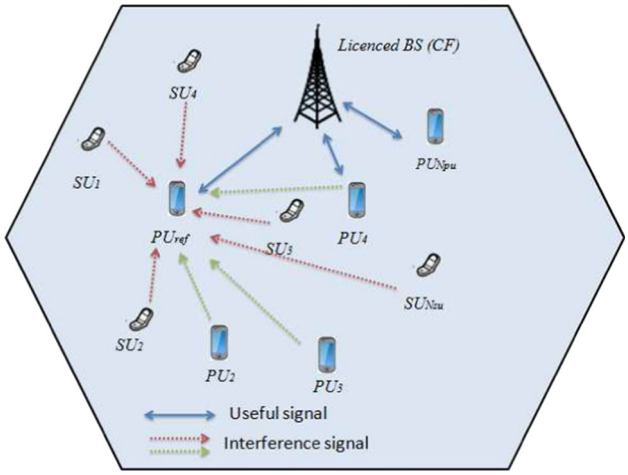


**Fig. 1** Cognitive radio network

(3) Compare and evaluate the accuracy of the theoretical approach for both OFDM and FBMC considering PHYDYAS prototype filters.

The PU is using his licensed channel which is divided into time slots. When the PU does not use some of his spectrum band, it leases some of its spectrum access time to the SU for a period of time. The reference mobile user is located at $(u, v)$; the reference base station is assumed to be situated at the origin $(u_0, v_0) = (0, 0)$.

The reference mobile is considered as a PU. The impulse response of multi-path channel between the $k$th BS and the reference mobile station (PU) is presented as follows [13],

$$h_k(t) = \sum_{i=0}^{L-1} h_{k,i}\delta\left(t - \frac{n_{k,i}}{N}T\right) \tag{1}$$

where $T$ is the symbol duration, $N$ is the total number of sub-carriers, $L$ is the total number of nodes, $n_{k,i}$ are the delay spreads of the channel $n_{k,0} \leq n_{k,1} \leq \cdots \leq n_{k,L-1} \leq C$, C is the maximum delay spread of the channel normalized by the sampling period $(\frac{T}{N})$, and $h_{k,i}$, are the complex channel path gains, which are assumed to be mutually independent [13].

In the experiment, we chose to focus on the block sub-carrier assignment schemes [13]. We assume a single block of adjacent sub-carriers is allocated to each cell and that $\delta$ sub-carriers $(\delta = 0, 1, \ldots)$ adjacent sub-carriers to each block are free and serve as guard bands between the different blocks. We also suppose that the propagation channels are stationary over one OFDM/FBMC symbol. Furthermore, the channel model includes path-loss effects. As a consequence, we can express the composite desired signal at the receiver as follows [14]:

$$r(t) = d_0^{-\beta} s_0(t) * h_0(t) \tag{2}$$

where $d_0$ is the distance between the reference PU and the $k$th node, and $\beta$ is the path loss exponent [18].

A list of the major notations used in this paper is given in Table 1.

### 3.1 Asynchronism in multi-carrier systems

Generally, in order to preserve the orthogonality between the sub-carriers in OFDM, the signals of the various users must arrive at the input of the receiver in a synchronous manner. Several synchronization techniques have been proposed in OFDM to ensure the alignment of signals at a receiver. Most of these techniques must be controlled in a closed loop [19], which requires estimation of the different arrival times by the receiver. Hence, it is important to take into account the harmful effect of interference caused by imperfect synchronization for multi-carrier based CR networks as it was considered in the system model depicted in Fig. 1.

### 3.2 Cognitive radio considerations

Since cognitive radio (CR) is performing all of the Spectrum sensing, spectrum management, spectrum sharing and spectrum mobility [20], we consider a cellular network operating in a wide band, in which the base station has the role of a coordination function (fusion center, FC), which principle role is spectrum sensing. The term 'cognitive radio' has evolved over the years to make the radio more capable and more powerful. We also classify CR into overlay or known interference models and underlay or interference avoidance models. It is a paradigm for wireless communication in which either a network or a wireless node changes its transmission or reception parameters to communicate efficiently while avoiding interference with licensed or unlicensed users. An important requirement of the CR network is to sense the spectrum holes and be aware of its surroundings. The proposed CR system operates as follows: First, the frequency band's status (active/idle) is

**Table 1** List of notations

| B | Total bandwidth |
|---|---|
| $f_c$ | Center frequency |
| T | Symbol duration |
| Nbloc | Number of blocks/frame |
| N | Number of sub-carriers |
| CP | Cyclic Prefix |
| $\tau$ | Timing offset |
| $\varphi$ | Phase offset |

checked in the first spectrum sensing. Second, when idle, the secondary transmitter concludes the frame and transmits information to the secondary receiver. At the end of the frame, if the presence of primary users is detected, data transmission will be ceased, so that PUs aren't affected by interference [21]. In the opposite case, the SU will access the frequency band again in the next frame. Finally, the process is repeated. An important requirement of the CR network is to sense the spectrum holes and be aware of its surroundings. Since a CR network is considered as an opportunistic network where only spectrum holes will be used, the PU activity is a relevant issue for the SU's performance. Different PU activity models are presented in [17]. These models give information about spectrum use by PUs. Even though the spectrum is idle, it can not be guaranteed that the spectrum holes will be available for SU during the entirety of transmission. That's why the state of the PU could be predicted by knowing its activity model [22].

### 3.3 PU/SU activity modelling

Different PU/SU activity modeling may be used in order to deal with DSA performance. The PU/SU activity may be based on statistics, on measured data or on a poisson modelling [23]. In the latter type of modeling, the arrival of each user is independent and we assume that the transmission follows a poisson arrival process. Since, we considered in the Fig. 1 that the CF node is responsible of providing spectrum holes to he SUs, it will also fill the role of spectrum sensing and PU/SU activity following. The Fig. 2 presents the asynchronous proposed scheme used in our system model and this flowchart depicts the interactions between both PU and SU in the block including asynchronous interference. In our multi-carrier CR based scheme, we consider that PU/SU activity is following a poisson modelling [24].

### 3.4 OFDM/FBMC transmission chain

#### 3.4.1 OFDM

Conventional orthogonal frequency-division multiplexing (OFDM) is generally considered a more matched technique for CR physical layer. It has been probably the most successful multi-carrier modulation (MCM) scheme for the wired or wireless communication in the past two decades. Each sub-carrier is independently modulated using QPSK phase modulation [25].

The flow symbols Ci are defined from two bits by a constellation of QPSK modulation. The multi-carrier signal will be then amplified by the power amplifier. At the output, we get the signal which will be sent by the antenna. Inverse operations are performed in the reception [26]. In CR cellular networks, we deal also with offset parameter causing asynchrony and interference.

In OFDM system, bits are mapped to constellation symbols where the modulation and demodulation are, respectively, insured by the inverse fast Fourier transform (IFFT) and the fast Fourier transform (FFT) [6].

#### 3.4.2 FBMC

One important MCM method is the filter bank based system (FBMC), which transmit pulses that are more localized in time-frequency domain. The orthogonality between sub-carriers is provided by introducing a half symbol period delay between the in-phase and the quadrature components of each complex symbol. This technique is called offset QAM (OQAM) technique. There are two kinds of implementations of FBMC: the frequency spreading filter bank multi-carrier (FS-FBMC) and the poly-phase network filter bank multi-carrier (PPN-FBMC). For two adjacent sub-carriers, if we consider that the time delay T/2 is introduced into the imaginary part of the QAM symbols on one of the sub-carriers, then it is introduced into the real part of the symbols on the other one.

## 4 Suggested asynchronous interference analysis

We consider that the reference mobile user is perfectly synchronized with its base station but not necessarily synchronized with the other base stations. The other PUs ($N_{PU}$) and SUs ($N_{SU}$) are not necessarily synchronized with their BSs and between each other [8]. The composite signal at the reference receiver is then expressed as the sum of the desired signal coming from the reference base station and the interference signal coming from the surrounding nodes, which can be secondary ($S_{s_j}^{\tau_j}$) or primary ($S_{p_i}^{\tau_i}$) signals and the Additive White Gaussian Noise $(AWGN)$ ($n(t)$). Therefore, we can write the total received signal at the reference PU of our system model, as described by the Eq. (3),

$$r(t) = \sum_{i=1}^{N_{PU}} S_{p_i}^{\tau_i}(t) + \sum_{j=1}^{N_{SU}} S_{s_j}^{\tau_j}(t) + n(t) \tag{3}$$

### 4.1 Asynchronous OFDM interference analysis

As we consider the asynchronous case, other transmitters are operating asynchronously in a wireless environment conditional to path loss and multipath fading. Then, we can easily deduce the following equation from Eq. (3):

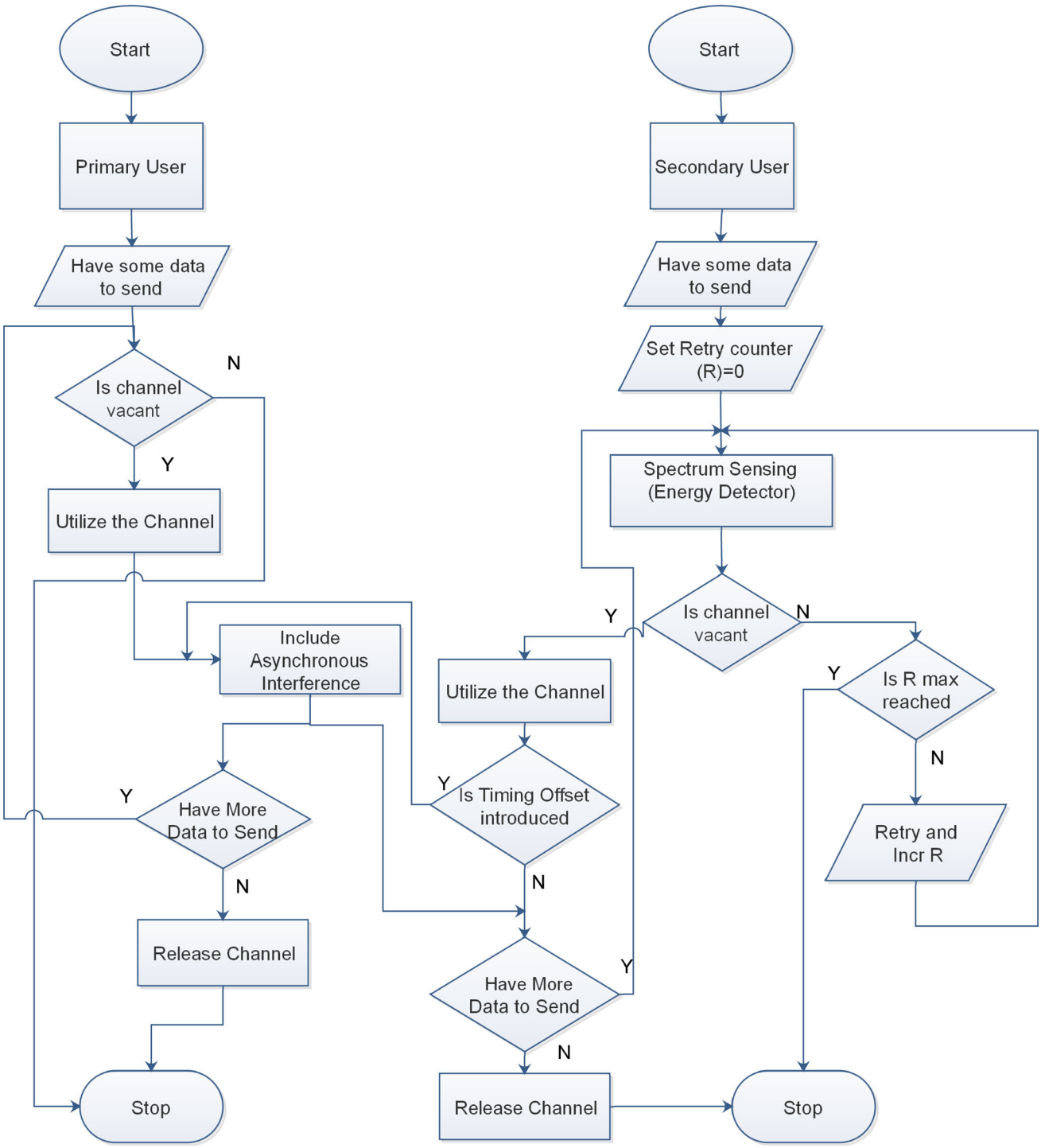


**Fig. 2** Asynchronous cognitive radio flowchart

$$r(t) = d^{-\beta} s_{PU_1}(t) h_1(t) + \sum_{i=2}^{N_{PU}} d^{-\beta} s_{PU_i}(t - \tau_i) h_i(t) + \sum_{j=1}^{N_{SU}} d^{-\beta} s_{SU_j}(t - \tau_j - \Delta_{sj}) h_j(t) + n(t) \quad (4)$$

The equation above denotes respectively the desired signal, the interference signal coming from PUs, the interference signal coming from SUs and the AWGN noise. And

– $\tau_i$ denotes the timing offset between the reference node and the $i$th PU,

- $\tau_j$ denotes the timing offset between the reference node and the $j$th SU ,
- $\Delta_{sj}$ denotes the sensing time used by the FC which could be considered the same for all users.
- $\beta$ path loss exponent.

Due to the timing offset between the neighboring cells and the reference one (where the the reference PU and SU exist), signals coming from other cells and cognitive network may be non orthogonal and will then generate asynchronous interference and cause a degradation in SINR.

Our theoretical study begins by analytically calculating the interference tables, in order to compute the bit error rate (BER) [9]. Underlay CR paradigm is used in our context. Therefore, only spectrum holes are used under the condition of the interference threshold and only one secondary user is transmitting over all available spectrum holes. The signal corresponding to the secondary user waveform is using OFDM multi-carrier modulation [9].

Consider the transmission of a single complex symbol x, on the $k$th frequency slot, from the interfering BS.

$$s(t) = x_k g(t-\tau_k) e^{j\frac{2\pi}{T}k(t-\tau_k)+j\varphi_k} \tag{5}$$

Our proposed model presents the interference signal corresponding from the SUs, we can easily write its expression as in Eq. (6):

$$S_s(t) = \sum_{j=1}^{N_{SU}} s_{s_j}(t) = \sum_{j=1}^{N_{SU}} \sum_{k'=1}^{F_{SU}} x_{k'} g(a_j) e^{j\frac{2\pi}{T}k' a_j} \tag{6}$$

where $g(t)$ denotes the transmit pulse shape having the form as in Eq. (7), $x_{k'}$ is the complex transmitted symbol, $F_{SU}$ denotes the total number of sub-carriers of all spectrum holes and $a_j = t - \tau_j - \Delta_{sj}$ . T and $\Delta$ represents respectively the symbol duration and the cyclic prefix (time unit).

$$g(t) = \begin{cases} \frac{1}{\sqrt{T}} & \text{if} \quad t \in [\Delta_s, T+\Delta+\Delta_s] \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

Supposing now, we have only one PU and one SU, the asynchronous OFDM interference signal received on the $k$th (PU frequency slot) sub-carrier for a single complex symbol $x_{k',0}$ on the $k'$th ( SU frequency slot/spectrum hole) sub-carrier is given by:

$$y_{k',PU_{ref}}(\tau_{SU_1}, \varphi_{SU_1}) = \left\langle s_{SU_1}(t-\tau_{SU_1}, \varphi_{SU_1}), f(t) e^{-j\frac{2\pi}{T}kt} \right\rangle \tag{8}$$

The receiver filter impulse response, f(t), is expressed exactly as the transmit pulse shape but when $t \in [\Delta_s, T+\Delta_s]$ ) and $l = k - k'$

Then, for $t = T + \Delta + \Delta_s$ we can deduce from (8) the following equation:

$$y_{k',PU_{ref}}(\tau_{SU_1}, \varphi_{SU_1}) = x_{SU}\alpha_{k'} \int_{-\infty}^{+\infty} g(t-\lambda_1) f(\lambda_2 - t) e^{j\frac{2\pi}{T}lt} dt \tag{9}$$

The expressions of $\alpha_{k'}$, $\lambda_1$ and $\lambda_2$ are presented in the Appendix 2. The value of $\tau_{SU_1}$ determines the limits of the integral, and then we can obtain when $l \neq 0$, the equation below:

$$y_{k',PU_{ref}}(\tau_{SU_1}, l) = \begin{cases} x_{SU} u_{k'} \delta(l) & \text{if} \quad \tau_{SU_1} \in [\Delta_s, \Delta] \\ x_{SU} u_{k'} \rho_l \dfrac{sin\left(\frac{\pi l}{T} a_1\right)}{\pi l} & \text{if} \quad \tau_{SU_1} \in [inf, sup] \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

where inf, sup, $a_1$, $u_{k'}$, $\rho_l$ are also expressed in the Appendix 2 and $\delta(l)$ is the Kronecker delta function.

By following the same steps, we get the expression of the asynchronous interference when $l = 0$ :

$$y_{k'PU_{ref}}(\tau_{SU_1}, 0) = \begin{cases} x_{SU} u_{k'} & \text{if} \quad \tau_{SU_1} \in [\Delta_s, \Delta] \\ x_{SU} u_{k'} \dfrac{a'_1}{T} & \text{if} \quad \tau_{SU_1} \in [inf, sup] \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

In the Appendix 2, we express $a'_1$. In our suggested analysis, we after express the corresponding interference power of the specific case when only one PU and one SU are as follows ($l \neq 0$):

$$I(\tau_{SU_1}, l) = \left| y_{k',PU_{ref}}(\tau_{PU_1}, l) \right|^2 = \begin{cases} \delta(l) & \text{if} \quad \tau_{PU_1} \in [\Delta_s, \Delta] \\ \left| \dfrac{sin\left(\frac{\pi l}{T} a_1\right)}{\pi l} \right|^2 & \text{if} \quad \tau_{SU_1} \in [inf, sup] \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

$$I(\tau_{SU_1}, 0) = \left| y_{k',PU_{ref}}(\tau_{SU_1}, 0) \right|^2 = \begin{cases} 1 & \text{if } \tau_{SU_1} \in [\Delta_s, \Delta] \\ \left| \dfrac{a'_1}{T} \right|^2 & \text{if } \tau_{SU_1} \in [inf, sup] \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

Following [27], the resulting general interference power is the sum of interference powers coming from all other SU in reference to the primary reference user $PU_{ref}$ (see Fig. 1). We follow indeed the same steps as previously described single-user case. We have then expressed the

general interference coming only from secondary users referred to $PU_{ref}$ as follows:

**Case 1**: $\Delta_s \leq \tau_{SU_i} \leq \Delta$

$$I_{k_i',PU_{ref}}(\tau_{SU_i}, \varphi_{SU_i}, l_i) = \left| \sum_{i=1}^{N_{SU}} K_i \delta(l_i) \right|^2 \tag{14}$$

**Case 2**: $\Delta_s + \Delta \leq \tau_{SU_i} \leq T + \Delta + \Delta_s$

$$I_{k_i',PU_{ref}}(\tau_{SU_i}, \varphi_{SU_i}, l_i) = \left| \sum_{i=1}^{N_{SU}} U_i \frac{sin\left(\frac{\pi l_i}{T} a_i\right)}{\pi l_i} \right|^2 \tag{15}$$

**Case 3**: $\tau_{SU_i} \geq T + \Delta + \Delta_s$

$$I_{k_i',PU_{ref}}(\tau_{SU_i}, \varphi_{SU_i}, l_i) = 0 \tag{16}$$

The expressions $K_i$, $U_i$, $a_i$ are also presented in the Appendix 2. The interference term coming from other PUs are calculated following the same method above, and the general interference term will be the sum of those of PUs and SUs.

### 4.2 Asynchronous FBMC interference analysis

The FBMC multicarrier modulation does not require any CP extension and can overcome the spectral lack problem by minimizing the side-lobes of each subcarrier and therefore lead to high efficiency (in terms of spectrum and interference) [13]. Since the transmission in FBMC system is based on the division of the bandwidth into a large number of uniform sub-bands, the idea of the offset QAM is to transmit a delayed QAM symbol, which reduce the inter-symbol interference and the Inter-carrier interference [9]. Throughout this section, we are going to present some details about how to model the asynchronous interference tables in multi-user CR network using FBMC (N PU and N SU). We will begin by presenting the single-user case in order to make our approach clearer.

The FBMC signal in time domain is presented by Eq. (17) as in [26]:

$$s(t) = \sum_{m=0}^{N-1} \sum_{n=-\infty}^{\infty} a_{m,n} \gamma_{m,n}(t) \tag{17}$$

with $\gamma_{m,n}(t)$ and $\varphi_{m,n}$ are presented as in Eqs. (18) and (19):

$$\gamma_{m,n}(t) = g\left(t - n\frac{T}{2}\right) e^{j\frac{2\pi}{T}mt} e^{j\varphi_{m,n}} \tag{18}$$

$$\varphi_{m,n} = \frac{\pi}{2}(m+n) - \pi mn \tag{19}$$

Since in our analysis, we have considered the PHYDYAS NPR (nearly perfect reconstruction) prototype filter using the frequency sampling technique [26], the impulse response of this filter is defined as follows:

$$g(t) = \begin{cases} \frac{1}{\sqrt{A}} \left[ 1 + 2\sum_{k=1}^{K-1} (-1)^k Y_k(t) \right] & \text{if} \quad t \in [0, KT] \\ 0 & \text{otherwise} \end{cases} \tag{20}$$

where K is the overlapping factor and A is the normalization factor [17] and $Y_k(t) = G_k cos(\frac{2\pi}{KT} kt)$.

In order to simplify expression and make then easier to manipulate, we suggest to define the following integral in Eq. (21). The explicit form expressions of this integral is provided in Appendix 1.

$$\psi(t, \tau, l)|_{t_1}^{t_2} = \int_{t_1}^{t_2} g(t-\tau) g(t) e^{j\frac{2\pi}{T}lt} \, dt \tag{21}$$

For the FBMC case, we refer to the same unsynchronized transmission chain, so we have expressed the composite received signal for $F_{PU}$, $F_{SU}$ sub-carriers as follows:

$$\begin{aligned} r(t) = {} & d^{-\beta} s_{PU_{ref}}(t) h_1(t) + \sum_{i=2}^{N_{PU}} d^{-\beta} s_{PU_i}(t - \tau_i) h_i(t) \\ & + \sum_{j=1}^{N_{SU}} d^{-\beta} s_{SU_j}(t - \tau_j - \Delta_{sj}) h_j(t) + n(t) \end{aligned} \tag{22}$$

With

$$s_{PU_{ref}}(t) = \sum_{m=1}^{F_{PU}} \sum_{n=-\infty}^{+\infty} a_{m,n} \gamma_{m,n}(t) \tag{23}$$

$$s_{SU_j}(t) = \sum_{m'=1}^{F_{SU}} \sum_{n'=-\infty}^{+\infty} b_{m',n'} \gamma_{m',n'}(t - \tau_j - \Delta_{sj}) e^{j\varphi} \tag{24}$$

Now let's consider the asynchronous transmission [9] of secondary users same as we suppose the primary user as the reference of our analytic approach. In this section, we have expressed the FBMC interference signal for a constant unitary channel. Namely, interference can be considered as the interference signal coming from a single path. The $m_0$-th output of the receiver filter on the $n_0$-th signalling interval.

$$y_{m_0,n_0}(\tau_{SU_1}, \varphi_{SU_1}) = \langle s_{SU_1}, \gamma_{m,n} \rangle \tag{25}$$

By substituting the Eqs. (23) and (24), we get then the $m_0$th output of the receiver filter on the $n_0$th signaling interval (i.e $t = n_0 T/2$) coming from $s_{SU_1}(t - \tau, \varphi)$ :

$$\begin{aligned} y_{m_0,n_0}(\tau, \varphi) = {} & \sum_{m'=1}^{F_{SU}} \sum_{n'=-\infty}^{+\infty} b_{m',n'} e^{j\varphi} e^{-j\frac{2\pi}{T}m'\tau} e^{-j(\varphi_{m_0,n_0} - \varphi_{m',n'})} \\ & \psi(t, \tau', l)|_{t=-\infty}^{+\infty} \end{aligned} \tag{26}$$

To make it simplified, we defined the expression of $\psi(t,\tau,l)$ same as in Eq. (23), $l$ and $\tau'$ in Appendix 1 and 2. We have to consider two cases:

**Case 1**: $\tau \geq (n'-n_0)\frac{T}{2}$

$$y_{m_0,n_0}(\tau,\varphi) = \sum_{m'=1}^{F_{SU}} \sum_{n'=-\infty}^{+\infty} b_{m',n'} e^{j\phi} e^{-j\frac{2\pi}{T}m'\tau} e^{-j\phi'} \psi(t,\tau',l)\big|_{t=\tau+(n'-n_0)\frac{T}{2}}^{KT} \tag{27}$$

**Case 2**: $\tau \leq (n'-n_0)\frac{T}{2}$

$$y_{m_0,n_0}(\tau,\varphi) = \sum_{m'=1}^{F_{SU}} \sum_{n'=-\infty}^{+\infty} b_{m',n'} e^{j\phi} e^{-j\frac{2\pi}{T}m'\tau} e^{-j\phi'} \psi(t,\tau',l)\big|_{t=0}^{KT+\tau+(n'-n_0)\frac{T}{2}} \tag{28}$$

The expression of $\phi$ and $\phi'$ are presented in Appendix 2. Then, after OQAM decision, we can write the total complex symbol as in Eq. (29):

$$y_{tot}(\tau,\varphi) = Re(y_{m_0,n_0}(\tau,\varphi)) + jRe(y_{m_0,n_0}(\tau,\varphi)) \tag{29}$$

And the corresponding interference power table can be given through the following expression:

$$I(\tau,l) = E\left[|y_{tot}(\tau,\varphi)|^2\right] \tag{30}$$

Now let's consider the general case where multiple users are present. We can express the signal of the reference primary user and of each secondary user respectively as in Eqs. (31) and (32).

$$s_{PU_{ref}}(t) = \sum_{m=1}^{F_{PU}} \sum_{n=-\infty}^{+\infty} a_{m,n}\gamma_{m,n}(t) \tag{31}$$

$$s_{SU_j}(t) = \sum_{m'=1}^{F_{SU}} \sum_{n=-\infty}^{+\infty} b_{m',n'}\gamma_{m,n}(t-\tau_j-\Delta_{sj})e^{j\varphi} \tag{32}$$

If we consider the asynchronous transmission of two secondary users and two primary ones, analytic development will be in reference to one primary user.

$$y_{m,n}(\tau_{SU_i},\varphi_{SU_i}) = \langle s_{PU_1} + s_{SU_1} + s_{SU_2}, \gamma_{m,n}\rangle \tag{33}$$

After some mathematical development of the equation above we get then:

$$y_{m,n}(\tau_{SU_i},\varphi_i) = \sum_{i=1}^{N_{SU}} \sum_{m'_j=1}^{F_{SU}} \sum_{n'=-\infty}^{+\infty} b_{m'_j,n'} e^{j\varphi_j} e^{-j\frac{2\pi}{T}m'_i\tau_{SU_i}} e^{-j(\varphi_{m_0,n_0}-\varphi_{m'_i,n'})} \psi(t,\tau'_i,l_i)\big|_{t=-\infty}^{+\infty} \tag{34}$$

$l_j$ and $\tau'_i$ are also expressed in the Appendix 2. By using the same steps as in the single-user case, we can easily express the following equations:

**Case 1**: $\tau_{SU_i} \geq (n'-n_0)\frac{T}{2}$

$$y_{m_0,n_0}(\tau_i,\varphi_j) = \sum_{i=1}^{N_{SU}} \sum_{m'_j=1}^{F_{SU}} \sum_{n'=-\infty}^{+\infty} b_{m'_j,n'} e^{j\phi_i} e^{-j\frac{2\pi}{T}m'_j\tau_j} e^{-j(\phi'_i)} \psi(t,\tau'_j,l)\big|_{t=\tau_j+(n'-n_0)\frac{T}{2}}^{KT} \tag{35}$$

**Case 2**: $\tau_j \leq (n'-n_0)\frac{T}{2}$

$$y_{m_0,n_0}(\tau_j,\varphi_j) = \sum_{j=1}^{N_{SU}} \sum_{m'_j=1}^{F_{SU}} \sum_{n'=-\infty}^{+\infty} b_{m'_j,n'} e^{j\phi_i} e^{-j\frac{2\pi}{T}m'_j\tau} e^{-j\phi'_i} \psi(t,\tau'_j,l)\big|_{t=0}^{KT+\tau_j+(n'-n_0)\frac{T}{2}} \tag{36}$$

$\phi_i$ and $\phi'_i$ are expressed in Appendix 2. As it was previously done, after OQAM decision, we can write the total complex symbol

$$y_{tot}(\tau_j,\varphi_j) = Re(y_{m_0,n_0}(\tau_j,\varphi_j)) + jRe(y_{m_0,n_0}(\tau_j,\varphi_j)) \tag{37}$$

and the corresponding interference power table can be given by the following expression:

$$I(\tau_j,l_j) = E\left[\left|y_{tot}(\tau_j,\varphi_j)\right|^2\right] \tag{38}$$

### 4.3 Bit error rate analysis

It was developed in [8] that the asynchronous interference power arriving through a frequency selective channel can be calculated as expressed in Eq. (39), in the case of multiuser CR network.

We present BER performance analysis in the presence of multiple PU and SU. By following the same steps as in the single-user case,we express the asynchronous interference power as in Eq. (39):

$$P_{interf}(m) = \upsilon_{pu} + \upsilon_{pu} \tag{39}$$

where

$$\upsilon_{pu} = \sum_{i=2}^{N_{PU}} \sum_{m\in F_{PU}} d_i^{-\beta} P_{trans}(m) I(\tau_i,|m-m'|)|H_i(m')|^2$$

$$\upsilon_{SU} = \sum_{j=1}^{N_{SU}} \sum_{m'\in F_{SU}} d_j^{-\beta} P_{trans}(m') I(\tau_j,|m-m'|)|H_j(m')|^2$$

where $F_{PU}$, $F_{SU}$ are the set of sub-carriers assigned respectively to PU and SU, $\tau_i$, $\tau_j$ are the timing offset of respectively PU and SU, $H_i(m')$, $H_j(m')$ are the power channel gain between the PU reference user for respectively the $i$th PU and $j$th SU

We now derive new expressions for the BER [28], in the presence of asynchronous interference for OFDM/FBMC cognitive radio networks. In our study, we present the theoretical expression of SINR as in Eq. (40):

$$SINR(m) = \frac{d_1^{-\beta} P_{trans}(m)|H_0(m)|^2}{U + V + N_0 B0} \tag{40}$$

then we obtain :

$$SINR(m) = \frac{|H_0(m)|^2}{\chi_{su} + \chi_{pu} + b} \tag{41}$$

where

$$\chi_{PU} = \sum_{i=2}^{N_{PU}} \sum_{m' \in F_{PU}} A_{i,m,m'} |H_i(m')|^2$$

$$\chi_{SU} = \sum_{j=1}^{N_{SU}} \sum_{m' \in F_{SU}} A_{j,m,m'} |H_j(m')|^2$$

$U$, $V$, $A_{i,m,m'}$, $A_{j,m,m'}$, $b$ are expressed in Appendix 2. In the following derivation, we analyze the M-QAM constellation case. In the literature, the BER of M-QAM modulation [26, 29] in Rayleigh fading channel is given in the Eq. (42):

$$BER_{MQAM} = \frac{(M-1)}{Mlog_2 M}\left[1 - \frac{1}{\sqrt{1 + \frac{M^2-1}{3*SINR*log_2 M}}}\right] \tag{42}$$

## 5 Results and discussion

In this section, we verify the correctness and the validity of the proposed method by means of numerical results that illustrate the impact of asynchrony. In the context of CR, we assume that some licensed spectrum band resources are already obtained by CR spectrum sensing.

In order to evaluate the effects of interference caused by the imperfect synchronization in multi-carrier techniques, we consider the system model depicted in Fig. 1. The reference receiver is assumed to be perfectly synchronized with its corresponding transmitter. Moreover, the timing offset of each user $\tau$ and the phase offset of each user $\varphi$ are assumed to be uniform random variables that are distributed on $\tau$ in [0, T] and $\varphi$ in $[0, 2\pi]$, respectively.

Our theoretical approach results are presented for different values of timing offset and by varying the number of secondary users. In Figs. 3 and 4, we present the interference level in CR networks in the single-user case ($N_{SU} = 1$) for both OFDM and FBMC cases. We plot the asynchronous interference for three values of timing offset $\tau = \frac{T}{4}, \frac{T}{3}, \frac{T}{2}$ in order to compare the performance between both multi-carrier schemes. Indeed, the comparison between the two theoretical curves confirm that FBMC is best suited for CR networks, which confirms also the precision of the analytic approach for single case. Our

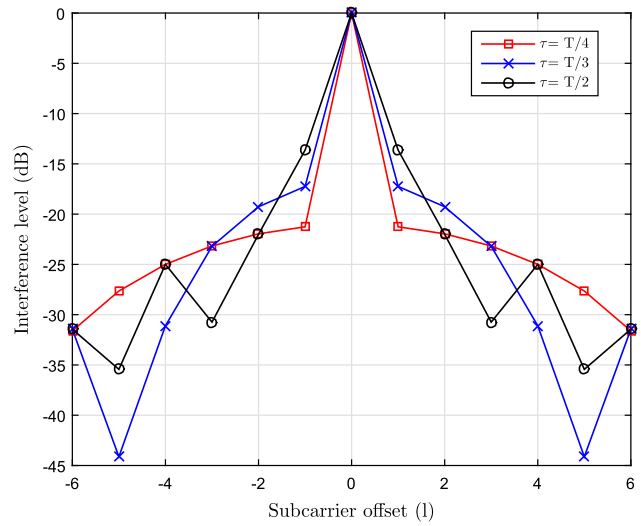


**Fig. 3** The interference level in OFDM single-user CR network for $\tau = \frac{T}{4}, \frac{T}{3}, \frac{T}{2}$

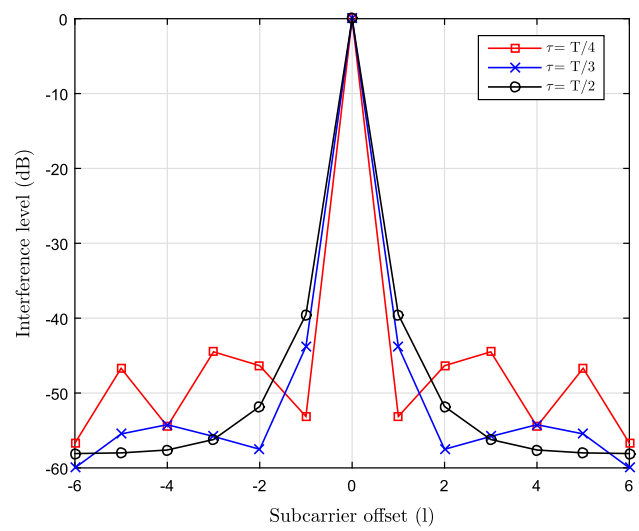


**Fig. 4** The interference level in FBMC single-user CR network for $\tau = \frac{T}{4}, \frac{T}{3}, \frac{T}{2}$

simulation results and the interference analysis implementation are obtained using Matlab tool, simulation criteria are given in Table 4.

In Tables 2 and 3, we give some values of the instantaneous interference tables (single-user case) when using both multicarrier techniques (OFDM/ FBMC) for $\tau = \frac{T}{2}, \frac{T}{3}, \frac{T}{4}$. These examples are also depicted in Figs. 5 and 6 where the interference level varies with respect to the timing offset $\tau$ and the spectral distance l.

Using the expressions given in Sect. 3, we plot Figs. 5 and 6, which show the interference power caused by different timing offsets $\tau$ for the multi-user case. We have also

**Table 2** OFDM instantaneous interference tables (single-user case) (dB) for $\tau = \frac{T}{2}, \frac{T}{3}, \frac{T}{4}$

| $l$ | $\tau = \frac{T}{2}$ | $\tau = \frac{T}{3}$ | $\tau = \frac{T}{4}$ |
|---|---|---|---|
| $l = 0$ | 0 | 0 | 0 |
| $l = 2$ | − 22.01 | − 19.26 | − 22 |
| $l = 4$ | − 25 | − 31.10 | − 25 |
| $l = 6$ | − 31.44 | − 31.44 | − 31.60 |

**Table 3** FBMC instantaneous interference tables(Mono-User Case) for $\tau = \frac{T}{2}, \frac{T}{3}, \frac{T}{4}$

| $l$ | $\tau = \frac{T}{2}$ | $\tau = \frac{T}{3}$ | $\tau = \frac{T}{4}$ |
|---|---|---|---|
| $l = 0$ | 0 | 0 | 0 |
| $l = 2$ | − 51.8 | − 57.51 | − 44.35 |
| $l = 4$ | − 57.66 | − 54.23 | − 57.45 |
| $l = 6$ | − 58.08 | − 60 | − 56.7 |

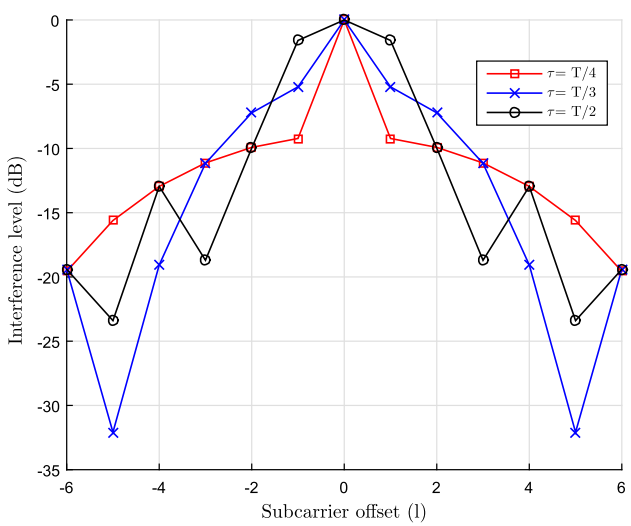


**Fig. 5** The interference level in OFDM multi-user case for $\tau = \frac{T}{4}, \frac{T}{3}, \frac{T}{2}$

considered a CP duration of $\frac{T}{8}$. It is worth noting that we can model this interference power as a set of instantaneous interference tables for each timing offset $\tau$. In figures above, we present the asynchronous interference term coming from secondary user in reference to the $PU_{ref}$.

The interference is inversely proportional to the number of SUs. It is noticed that, even though the number of SUs is increased, the use of FBMC scheme is still better than OFDM in terms of asynchronous interference. These results prove the effectiveness of the use of FBMC in CR networks and the accuracy of our theoretical analysis. As shown in the Figs. 3 and 5, for $\tau = \frac{T}{4}$, OFDM displays

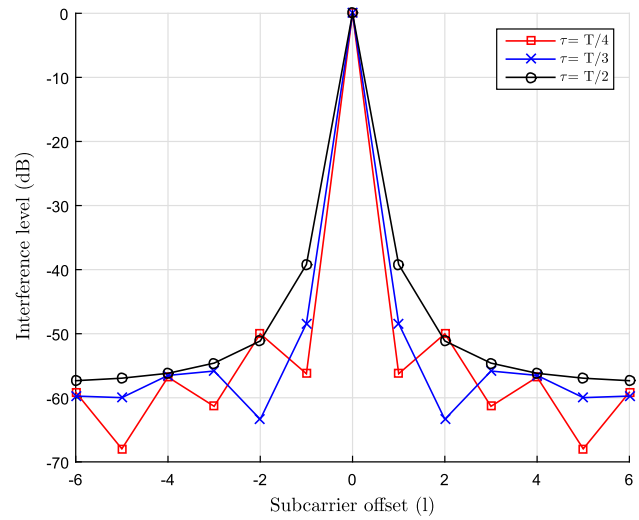


**Fig. 6** The interference level in FBMC multi-user case for $\tau = \frac{T}{4}, \frac{T}{3}, \frac{T}{2}$

substantial difference in normalized interference level between the single-user case (− 20 dB) and multi-user case (− 32 dB). Same tests conducted with FBMC technique show that the differences between the two cases is barely perceived (both values are approximately equal to − 55 dB).

In this part, we examine the influence of timing misalignment on the BER of an OFDM/FBMC system [30]. We consider a system with N = 1024 sub-carriers where the sampling frequency is 10 MHz. The noise term is characterized as an Additive White Gaussian Noise. Finally, for the Rayleigh fading channel, we have considered the pedestrian-A model with relative delays [0 110 190 410] ns and corresponding average powers [0 − 9.7 − 19.2 − 22.8] dB [30], the rest of system parameters are given in Table 4. In their analysis, authors in [14] have presented the effect of asynchronism on the system's performance for a non CR based multi-carrier cellular network. Our results were compared to those found in their work and it is also clear that CR technique increases the system's performance.

**Table 4** System simulation parameters

| B | 10 MHz |
|---|---|
| $f_c$ | 2.5 GHz |
| T | 102.4e−6 s |
| Nbloc | 100 |
| N | 1024 |
| CP | T/8 |
| Modulation | MQAM |
| $\tau$ | [0 T] |
| $\varphi$ | [0 2π] |

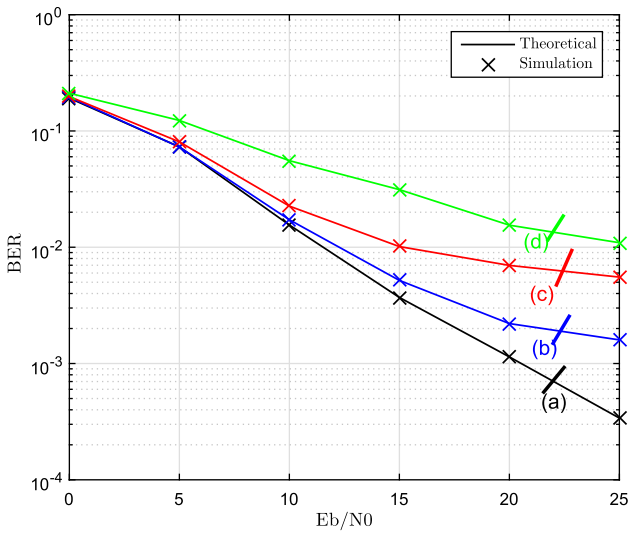


**Fig. 7** The OFDM average BER against the Eb/N0 for $\tau \in [0, T]$: **a** linear and perfectly synchronized (PS) scenario, **b** CR asynchronous 1PU–1SU scenario, **c** CR asynchronous 1PU-2SU scenario, **d** Non-CR asynchronous OFDM based cellular scenario

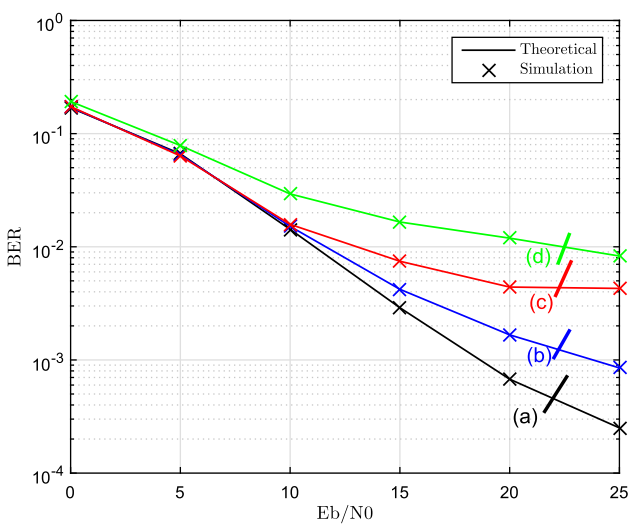


**Fig. 8** The FBMC average BER against the Eb/N0 for $\tau \in [0, T]$: **a** linear and perfectly synchronized (PS) scenario, **b** CR asynchronous 1PU–1SU scenario, **c** CR asynchronous 1PU–2SU scenario, **d** non-CR asynchronous FBMC based cellular scenario

In Figs. 7 and 8, we present the BER against Eb/N0 with Perfect Synchronization (PS) by considering asynchronous CR approach using OFDM and FBMC multi-carrier technique. The system performance for PS case and CR is the same until EB/N0 is equal to 10 dB for both multi-carrier techniques. The curves show that BER is directly proportional to the number of SUs (red and blue curves), which is expected since the asynchronous interference between the reference PU and SUs diminishes the system performance. Nevertheless, it is clear that our system's performance is higher than the BER rate of the system described in [14], which provides several reasons for the use of CR. The simulation results show that FBMC is best suited for CR networks since it provides BER improvements compared to OFDM, both of them confirm the validity of our theoretical approach.

## 6 Conclusion and future directions

In this paper, we have studied and developed a new theoretical aspect of interference analysis in a CR networks using OFDM/FBMC. In fact, this study is valid for analysing the impact of the asynchrony on OFDM/FBMC for a limited number of users (One PU, One SU), and for a multi-user case of $N_{PU}$ and $N_{SU}$, by estimating the interference introduced and establishing analytic expressions of the SINR. We further explored simulation results and compared to analytic expressions in terms of BER.

Actually, presented numerical results show that when the secondary system is present, the asynchronous interference caused by sub-carrier misalignment affects the system's performance in term of BER. Furthermore, the actual values of interference are higher when using OFDM than those planned for PHYDAS/FBMC, which proves the fact that FBMC multi-carrier technique is best suited for 5G networks in term of QoS. In this work, we conclude that the use of CR does not affect asynchronous interference level while optimizing the spectrum band. For example, for $\tau = \frac{T}{4}$ and $1 \in [-6, 6]$, the OFDM normalized interference level does not exceed -20dB, which is roughly the same results obtained in the non CR case study [18].

Finally, in our future research, we plan to extend this work for the multi-user case and for other types of channel characteristics. We hope that further tests for system's performance evaluation will be done using the ROC curves. It would be interesting to develop a new sensing algorithm in the context of multi-carrier networks to deal with interference mitigation [31]. To further our research, we plan to investigate the application of Channel Bonding (CB) technique [32] in order to increase the spectral efficiency of an asynchronous multi-carrier CR network.

## Appendix 1

After integration, we obtain

$$\psi(t,\tau,0)|_{t_1}^{t_2} = \frac{t}{A}\left[1+2\sum_{n=1}^{K-1} G_n^2 cos\left(\frac{2\pi}{KT}n\tau\right)\right] + \frac{KT}{\pi A}\left\{\sum_{n=1}^{K-1}\sum_{m=1}^{K-1}(-1)^{n+m}\frac{G_nG_m}{n+m} sin\left(\frac{2\pi}{KT}\left((n+m)t-n\tau\right)\right) + \sum_{n=1}^{K-1}\sum_{m=1}^{K-1}(-1)^{n+m}\frac{G_nG_m}{n-m} sin\left(\frac{2\pi}{KT}\left((n-m)t-n\tau\right)\right) + \sum_{n=1}^{K-1}(-1)^n\frac{G_n}{n}\left[sin\left(\frac{2\pi}{KT}n\tau\right) sin\left(\frac{2\pi}{KT}n(t-\tau)\right)\right]\right\}\Bigg|_{t=t_1}^{t_2} \tag{43}$$

when $l \neq 0$,

$$\psi(t,\tau,l)|_{t_1}^{t_2} = \frac{T}{j2\pi lA}e^{j\frac{2\pi}{T}lt} + \frac{KT}{j2\pi A}\left\{\sum_{n=1}^{K-1}(-1)^n G_n\left[\frac{1+e^{-j\frac{2\pi}{KT}n\tau}}{n+KL}e^{j\frac{2\pi}{KT}(n+KL)t} - \frac{1+e^{j\frac{2\pi}{KT}n\tau}}{n-KL}e^{-j\frac{2\pi}{KT}(n-KL)t}\right] + \frac{KT}{j2\pi A} gt\sum_{n=1}^{K-1}\sum_{m=1}^{K-1}(-1)^{n+m}G_nG_m\left[e^{-j\frac{2\pi}{KT}n\tau}\left(\frac{e^{j\frac{2\pi}{KT}(n+m+Kl)t}}{n+m+KL} + \frac{e^{j\frac{2\pi}{KT}(n-m+Kl)t}}{n-m+KL}\right) - e^{j\frac{2\pi}{KT}n\tau}\left(\frac{e^{-j\frac{2\pi}{KT}(n+m-Kl)t}}{n+m-KL} + \frac{e^{-j\frac{2\pi}{KT}(n-m-Kl)t}}{n-m-KL}\right)\right]\Bigg|_{t=t_1}^{t_2} \tag{44}$$

## Appendix 2

Parameters of Eq. (9):

$$\alpha_{k'} = e^{-j\frac{2\pi}{T}k'\tau_{SU_1}+j\varphi_{SU_1}}$$
$$\lambda_1 = \tau_{SU_1} + \Delta_s$$
$$\lambda_2 = T + \Delta + \Delta_s$$

Parameters of Eq. (10):

$$inf = \Delta_s + \Delta$$
$$sup = \Delta_s + T + \Delta$$
$$a_1 = 2\Delta_s + \tau_{SU_1} - \Delta$$
$$u_{k'} = e^{-j\frac{2\pi}{T}k'(\tau_{SU_1}+\Delta_s)+j\varphi_{SU_1}}$$
$$\rho_l = e^{-j\frac{2\pi l}{T}(\Delta+\tau_{SU_1}+2\Delta_s)}$$

Parameters of Eq. (12):

$$a_1' = T + \Delta - 2\Delta_s - \tau_{SU_1}$$

Parameters of Eqs. (14), (15) and (16):

$$K_i = x_{SU_i}e^{-j\frac{2\pi}{T}k_i'(\Delta_s+\tau_{SU_i})+j\varphi_{SU_i}}$$
$$U_i = x_{SU_i}e^{-j\frac{2\pi}{T}k_i'(\Delta_s+\tau_{SU_i})+j\varphi_{SU_i}}e^{-j\frac{2\pi l_i}{T}(\Delta+\tau_{SU_i}+2\Delta_s)}$$
$$a_i = 2\Delta_s + \tau_{SU_i} - \Delta$$

Parameters of Eq. (26):

$$l = m' - m$$
$$\tau' = (n'-n_0)\frac{T}{2} + \tau$$

Parameters of Eqs. (27) and (28):

$$\phi = \varphi_i + \varphi_{m,n} - \varphi_{m_0,n_0}$$
$$\phi' = \varphi_{m_0,n_0} - \varphi_{m_j',n'}$$

Parameters of Eqs. (34):

$$l_j = m_j' - m$$
$$\tau_i' = (n'-n_0)\frac{T}{2} + \tau_{SU_j}$$

Parameters of Eqs. (35) and (35):

$$\phi_i = \varphi_i + \varphi_{m,n} - \varphi_{m_0,n_0}$$
$$\phi_i' = \varphi_{m_0,n_0} - \varphi_{m_j',n'}$$

Parameters of Eqs. (40) and (41):

$$U = \sum_{i=2}^{N_{PU}}\sum_{m'\in F_{PU}} d_i^{-\beta}P_{trans}(m')I(\tau_i, m-m'|)|H_i(m')|^2$$
$$V = \sum_{j=1}^{N_{SU}}\sum_{m'\in F_{SU}} d_j^{-\beta}P_{trans}(m')I(\tau_j, |m-m'|)|H_j(m')|^2$$
$$A_{i,m,m'} = \left[\frac{d_i}{d_1}\right]^{-\beta}\frac{P_{trans}(m')}{P_{trans}(m)}I(\tau_i, |m-m'|)$$
$$A_{j,m,m'} = \left[\frac{d_j}{d_1}\right]^{-\beta}\frac{P_{trans}(m')}{P_{trans}(m)}I(\tau_j, |m-m'|)$$
$$b = \frac{N_0B_0}{d_1{}^{-\beta}P_{trans}(m)}$$

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